Unleash your potential

Master

Presenter's Manual

For

Advanced Excel (MS Excel)

Curriculum Code: OV-1501

Effective from: March 2024
Ver 1.0

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	March 2024	-	-

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1. Introduction

At the end of this module, student will be able to:

- Explain customization of the Ribbon and Quick Access Toolbar
- Explain proofing and management of Add-ins
- Describe built-in and custom templates in Excel
- Explain importing and exporting of XML data in Excel
- Describe worksheet protection and locking/unlocking of cells
- Explain workbook protection, finalization, and encryption of workbook
- Explain shared workbook and change tracking in a shared workbook
- Explain the process of recovering unsaved versions of a file
- Explain tracing of formula precedents and dependants
- Explain Data Validation tool and checking common errors in a file
- Explain iterative calculation and setting calculation precision
- Describe the use of single cell and multi cell array formula
- Explain SUMIFS(), COUNTIFS(), and basic counting functions
- Describe Statistical and Financial functions
- Describe Date and Time functions
- Describe Text, Logical, and LookUp functions
- Explain the importance of chart and the different types of chart
- Explain modification of chart elements, chart area, plot area, and chart axes
- Describe formatting of chart title, legend, and gridlines
- Explain Trend lines and data series
- Explain Dual axes and chart templates
- Describe combination chart, error bars, and data tables
- Explain conditional formatting of data using data bars, color scales, and icon sets
- Explain Sparklines
- Describe sorting and filtering of labels and values in a PivotTable Report
- Describe the use slicer for a PivotTable
- Explain creation of PivotChart Report
- Explain changing of the chart type of a PivotChart Report
- Explain Analysis Tool Pak
- Explain single factor and two factor tool with and without replication
- Describe Correlation and Covariance
- Explain Exponential smoothing and F-Test two-sample for variances
- Describe Histogram and moving average
- Explain Random number generation and t-Test
- Describe Goal seek and Solver for what-if analysis
- Explain one-input and two-input data table and scenario
- Explain creating and running a macro
- Explain assigning macro to a command button
- Explain creation of a custom macro button on the Quick Access Toolbar
- Describe macro security
- Describe Form controls
- Work with Date, Time, Data Type and Conversion Functions
- Work with Financial Functions
- Link Workbooks and Summarize Data

2. Information on Session Allocation

Module	Online Hours	Skill Code
EXL	24	1296

EXL: Data Analysis with MS Excel

3. Module Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

- **Deliverables for Students:**

Module	Deliverable
Data Analysis with MS Excel	Kit Code & Name: **OV-1501-KIT01-01-A & Kit of 1 title for OV-1501 Advanced Excel** Book Code & Name: **AINNDAEXL10723E000 & Data Analysis with MS Excel**

- **Deliverables for Faculty:**

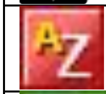
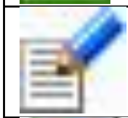
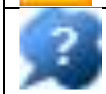
Module	Deliverable
Data Analysis with MS Excel	Kit Code & Name: **OV-1501-KIT01-01-A & Kit of 1 title for OV-1501 Advanced Excel** Book Code & Name: **AINNDAEXL10723E000 & Data Analysis with MS Excel**
Course	**Master PM** **Classroom Presentations** **Trainer Guide**

To aid the teaching process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description / Functionality
	***Download Book*** **-** Student has the option to download the subject related e-book and read offline
	***Glossary*** **-** Student can access a list of subject related specialized words with their definitions.
	***FAQ*** **-** Student can access frequently asked questions and their answers
	***Practice 4 Me*** **-** Student can test and evaluate their understanding of module related topics.
	***Work Assignments*** **-** Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	***References*** **-** Student can access additional subject related material for reading.
	***Feedback*** **-** Student can provide feedback on the course material
	***Ask to Learn*** **–** Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- **A Session is of 2 hours duration.**

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 EXL – TL1	Session 2 EXL – TL2	Session 3 EXL – TL3	Session 4 EXL – TL4
2	Session 5 EXL – TL5	Session 6 EXL – TL6	Session 7 EXL – TL7	Session 8 EXL – TL8
3	Session 9 EXL – TL9	Session 10 EXL – TL10	Session 11 EXL – TL11	Session 12 EXL – TL12

TL – Online Session

5. Session Coverage

The table below details out the contents of each session where:

Sess No.	Session Title	Session Details	Deliverables' Mapping
1 & 2	**EXL – TL1 & TL2**	All the topics as listed below from Session 1, Session 2, and Session 3 of *Data Analysis with MS Excel* book should be covered in this session. **Session 1 – Advanced Tools and Features in Excel** ➢ Understanding Add-ins ➢ Creating and Using Charts ➢ Working with Macros ➢ Linking Workbooks **Session 2 – Data Validation Using Functions** ➢ Built-in Data Validation Functions for Numbers and Text ➢ Customized Rules for Data Validation ➢ ISBLANK() ➢ ISNUMBER() ➢ ISTEXT() ➢ Lookup Functions with Data Validation **Session 3 – Performing Conditional Formatting** ➢ Built-in Rules to Highlight a Cell or a Cell Range ➢ Customized Conditional Formatting Rules ➢ Importing and Exporting XML Data	**Data Analysis with MS Excel** SG – Session 1,2, and 3 XP – Session 1,2, and 3 TG – Session 1, 2, and 3
3	**EXL – TL3**	The **Try It Yourself** questions from Session 1 to Session 3 of *Data Analysis with MS Excel* book should be covered in this session.	**Data Analysis with MS Excel** Session 1, 2, and 3
4 & 5	**EXL –TL4 & TL5**	All the topics as listed below from Session 4 and Session 5 of Data Analysis with MS Excel book should be covered in this session. **Session 4 – Data Manipulation Using Functions**	**Data Analysis with MS Excel** SG – Session 4 and 5 XP – Session 4 and 5 TG – Session 4 and 5

Sess No.	Session Title	Session Details	Deliverables' Mapping
		➢ Functions to Rank, Sort, Filter, and Transpose data ➢ Conversion Functions ➢ TEXT() ➢ VALUE() ➢ DOLLAR() ➢ FIXED() ➢ DATEVALUE() **<u>Session 5 – Summarizing Data Using Functions</u>** ➢ Mathematical and Trigonometric Functions ➢ Simplifying Calculations on Data ➢ Understanding Data Aggregation ➢ Summarizing Data with Database Functions ➢ DSUM() ➢ DMAX() ➢ DCOUNT() ➢ DAVERAGE()	
6	**EXL –TL6**	The **Try It Yourself** questions from Session 4 and Session 5 of *Data Analysis with MS Excel* book should be covered in this session.	**Data Analysis with MS Excel** Session 4 and 5
7 & 8	**EXL – TL7 & TL8**	All the topics as listed below from Session 6 and Session 7 of *Data Analysis with MS Excel* book should be covered in this session. **<u>Session 6 – Macros and Forms</u>** ➢ Introduction to Pivot Table ➢ Working with Pivot Table ➢ Introduction to Pivot Chart ➢ Working with Pivot Chart **<u>Session 7 – Data Analysis Using Statistical Functions</u>** ➢ Moving Average for a Given Time Period ➢ Frequency Distribution for the Given Observations ➢ Hypothesis Test for a Sample Mean ➢ Analyzing Data Using Functions ➢ Sampling ➢ Covariance ➢ Correlation	**Data Analysis with MS Excel** SG – Session 6 and 7 XP – Session 6 and 7 TG – Session 6 and 7

Sess No.	Session Title	Session Details	Deliverables' Mapping
		➢ Goal Seek ➢ Solver ➢ F-test ➢ Histogram	
9	**EXL – TL9**	The **Try It Yourself** questions from Session 6 and Session 7 of *Data Analysis with MS Excel* book should be covered in this session.	**Data Analysis with MS Excel** Session 6 and 7
10 & 11	**EXL – TL10 & TL11**	All the topics as listed below from Session 8 of *Data Analysis with MS Excel* 2016 book should be covered in this session. **<u>Session 8 – Working with Financial Functions</u>** ➢ ANOVA ➢ Regression Analysis ➢ Statistical Functions ➢ SLOPE() ➢ RSQ() ➢ INTERCEPT() ➢ TREND() ➢ Designing a Dashboard ➢ Creating and Using Charts for Dashboard	**Data Analysis with MS Excel** SG – Session 8 XP – Session 8 TG – Session 8
12	**EXL – TL12**	The **Try It Yourself** questions from Session 8 of *Data Analysis with MS Excel* book should be covered in this session.	**Data Analysis with MS Excel** Session 8

6. Library References

➢	**Data Analysis with Microsoft Excel** by Kenneth N. Berk
➢	**Business Analysis with Microsoft Excel** by Conrad Carlberg
➢	**Microsoft Excel Data Analysis and Business Modeling** by Curtis Frye

7. Classroom and Lab Pre-requisites

The requirements for conducting the sessions have been described as follows:

7.1 Software Requirements

Microsoft Windows 10 or higher
Microsoft 365

7.2 Recommended Hardware

Intel Core i3/i5 Processor or higher
8 GB RAM or above
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard

7.3 Other Requirements (For Classroom)

White board with marker
Computer for giving demonstrations wherever applicable
Computer / LCD projector for presenting the transparencies

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